AP PRECALCULUS

Unit 1 • Lesson 1.13 practice • Answer key

Lesson 1.13 • Function Model Selection and Assumption Articulation

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Data with equally spaced inputs have first differences $4.9,\ 5.1,\ 5.0,\ 4.9$. Which model is most appropriate?

(A) Linear

(B) Quadratic

(C) Cubic

(D) Rational

Solution

Answer: (A). The rate of change is roughly constant.

Question 2

NO CALCULATOR

Data with equally spaced inputs have second differences $-6.1,\ -5.9,\ -6.0$. Which model is most appropriate?

(A) Linear

(B) Quadratic

(C) Cubic

(D) Constant

Solution

Answer: (B). Roughly constant second differences indicate a quadratic model.

Question 3

NO CALCULATOR

A company's average cost per item is $A(n) = \dfrac{500 + 4n}{n}$ dollars, for n items. What does $\lim_{n \to \infty} A(n)$ tell you?

(A) The fixed cost is \$500.

(B) The average cost becomes 0.

(C) The average cost per item approaches \$4.

(D) The cost per item increases without bound.

Solution
Answer: (C). $A(n) = \frac{500}{n} + 4 \to 4$: the fixed cost is spread over more and more items.

Question 4

NO CALCULATOR

A quadratic model $P(t)$ gives the number of fish in a pond after t months; it has a maximum, then decreases. Which is the most reasonable restriction on the domain?

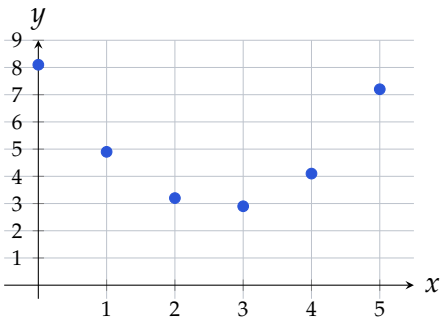
(A) All real numbers t
(B) $t < 0$
(C) Only integer values of P
(D) $t \geq 0$ and only while $P(t) \geq 0$

Solution
Answer: (D). Time starts at 0 and a population cannot be negative.

Question 5 (graph)

NO CALCULATOR

A scatter plot of data is shown. Which type of function is the most appropriate model for the data?



(A) Linear
(B) Quadratic
(C) Constant
(D) Rational with a vertical asymptote

Solution
Answer: (B). The data decrease, reach a single minimum, then increase, in a roughly symmetric way. That is the behavior of a quadratic function.

Question 6 (Free response)

NO CALCULATOR

A phone plan costs \$20 per month, which includes 100 texts. Each additional text costs \$0.10.

(a) Write a piecewise function $C(n)$ for the monthly cost of sending n texts.
(b) Find $C(250)$ and interpret it.
(c) For $n > 100$, the average cost per text is $A(n) = \dfrac{C(n)}{n}$. Find $\lim\limits_{n\to\infty} A(n)$ and interpret it.

Solution
(a) $C(n) = \begin{cases} 20, & 0 \leq n \leq 100 \\ 20 + 0.10(n - 100), & n > 100 \end{cases}$
(b) $C(250) = 20 + 0.10(150) = 35$: sending 250 texts in a month costs \$35.
(c) For $n > 100$, $A(n) = \dfrac{10 + 0.10n}{n} = \dfrac{10}{n} + 0.10 \to 0.10$. With many texts, the average cost per

> text approaches $0.10, the price of an extra text.